Reg. No: | 1 | 9 | 2 | 5 | 1 | 1 | 3 | 4 | 5 | | |

SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student : SoJenefar

Programme: B.E / B.Tech

Slot : A

Course Code: CSA1520

Faculty : Dr.Arul Raj A.M

Course Name: **Cloud Computing and Big Data Analytics**

Date: 02/11/2026

Duration: 1:30 Hr

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, S and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
	Total			50	

Student Signature	Faculty Signature	Program Director

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

1. Development in computing hardware

The development of computing hardware, such as powerful processors, increased RAM, high speed network and large storage device has made cloud computing possible.

Internet software technologies, virtualization, Web services and APIs allow application and resources to be accessed over the internet without maintaining physical infrastructure.

These development help organizations reduce cost, improve scalability and access computing resources whenever required.

2. Mapping requirement to cloud service model

(a) Iaas - Infrastructure as a service

used to move the existing content management application to virtual machines.

Provide virtual servers, storage and networking.

Example: Amazon EC2, Microsoft Azure VM

(a) Paas - Platform as a Service

used to develop & deploy new customer facing application

Provide development tools, runtime environment & databases

Example: Google App Engine, Azure App Service

c) Saas - Software as a Service

used to access a ready made CRM application through a web browser.

No need to install or maintain the software.

Example: Salesforce, Microsoft Dynamics 365

d) xaas - Everything as a Service

Provide IT resources and service through internet based on user requirments.

Example:

STaas (Storage as a service) = online storage

DBaas (Database as a service) = manage database

SECaas (Security as a service) = cloud based security

Naas (Network as a Service) = Network Services

Baas (Backup as a service) = cloud data backup

Benefits:

1. cost reduction
2. Scalability
3. Flexibility
4. Easy maintenance
5. High Availability

Conclusion:

Cloud computing evolution has enabled companies like stream wave to modernize their IT infrastructure. Iaas provide infrastructure control, Paas support application development and saas provide ready made software.

CONSOLIDATION AND ENERGY COST CALCULATION

Given Data

No of physical Servor : 15

Average CPU utilization : 25 %

Power consumption per old server : 350W

Power consumption per new host : 450 kl

Target CPU utilization per host : 65 %

Electricity cost : ₹ 9 per kwh

operating time : 24 x 365

Fault tolerance : N+1

(a) No of active host & total host including N+1

Total CPU Workload : 15 x 25% = 375 %

Active hosts required : $\frac{375}{65} = 5.77$

6 active hosts

Total host using N+1 : 7 hosts

(b) consolidation Ratio

consolidation : $\frac{\text{Old Server}}{\text{New host}} = \frac{15}{7}$

2.14 : 1

(c) CPU utilization

$\frac{375}{7 \times 65} \times 100 = 82.42\ \%$

After one host fail = $\frac{375}{6 \times 65} \times 100$

$= 96.15\%$

(d) Annual electricity cost saving

Old Server

$15 \times 350 = 5250 W = 5.25 kW$

New host

$7 \times 450 = 3150 W = 3.15 kW$

Annual operating hour

~~5.25~~ $24 \times 365 = 8760$ hrs

old annual

$5.25 \times 8760 \times 9 = ₹ 413910$

New annual

$3.15 \times 8760 \times 9 = ₹ 248346$

cost saving

$413910 - 248346$

₹ 1,65,564

IZATION HOST RESOURCES

(a) VCPU and memory over commitment

Total allocated VCPUS = 12 x 3 = 36 VCPUS

VCPUs = $\frac{36}{24}$ 1.5 : 1

Total allocated memory = 12 x 8 = 96 GB

Available memory = 96 - 8 = 88 GB

Memory ratio : 96/88 1.09 : 1

(b) Average and Peak memory sufficiency

Normal operation = 12 x 5 = 60 GB

Available memory = 88 GB

88 - 60 = 28 GB

Normal usage is 60 GB. So 28 is free

Peak operation : (8 x 8) + (4 x 6)

= 64 + 24 = 88 GB

88 - 88 : 0 GB

Peak memory usage is 88 GB, so no free

(c) Average & Peak CPU Sufficiency

Normal

36 x 0.35 = 12.6

Physical CPU core = 24

12.6 core used

CPU utilization

$$\frac{12.6}{24} \times 100 = 52.5$$

Peak operation:

$$86 \times 0.75 = 27$$

$$CPU = \frac{27}{24} \times 100 = 112.5\%$$

Q4) CLOUD FEDERATION, SDDC AND SOA

1. conventional cloud Architecture

- use physical server, Storage & network.
- Each university manage it own data centre
- Federation allow both university to share research and library resources.

2. SDDC

Manage resources through software.

Compute - CPU and memory allocation

Storage - Research data management

Networking - connect universities.

of SOA

- connect servie b/w universities
- uses REST APIs and SOAP
- Enable research and digital library sharing.

4. Challenge & Solution

challenges	Solution
Resource sharing	Resource Scheduling
Identify privacy	Federated identity
Unauthorized access	MFA
Data leakage	Encryption
Service failure	Backup & recovery
User Privacy	Access restriction

5. conclution

Cloud federation, SDDC and SOA enable universities to share resources. The federation challenges and privacy risks associated with user presence information and federated identity. It is suitable for techinal and administrative Safeguards.

Q5) CLOUD STORAGE, EDGE FILTERING AND SERVERLESS COMPUTING

Given data.

cameras = 150

video size = 4MB

one clip every 3 min

storage cost = ₹1.50 per GB-month

Serverless charge = ₹0.00015 per invocation

Edge filtering = 25% uploaded

1GB = 1000MB

(a) Daily data volume and bandwidth

volume

Clips per camera per day $= \frac{24 \times 60}{3} = 480$

Total clips = $150 \times 480 = 72000$

Daily data = $72000 \times 4 = 288000$ MB

= 288GB/per day

Average bandwidth = $\frac{288 \times 8}{86400}$

= 0.00267 Gbps

(b) Monthly storage cost without filtering

288 X30 = 8640 GB

8640 X 1.50 = ₹ 12,960

(c) Average Edge Filtering

Daily data : 288 X0.25 = 72GB

Average bandwidth : 0.00267 X 0.25

= 0.00067 Gbps

Monthly Storage = 72 X30

= 2160 GB

Storage cost = 2160 X1.50

= ₹ 3,240

12960 - 3240 = ₹ 9720

(d) monthly Serverless Invocation cost

Monthly invocation = 72000 X 30 X 0.25

= 540000

COST = 540000 X 0.000 15

= ₹ 81